Mohamed Salaheldeen

Curriculum Vitae

Institut de Mathématiques de Bourgogne (IMB), Dijon, France
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Education

Sep 2024 – May 2025 Dijon, France	**Institut de Mathématiques de Bourgogne (IMB)** M2 Research Master in Mathematical Physics Grade: 16.8/20 (ranked 1st) Thesis: First Contact with Factorization Homology Advisor: Prof. Lukas Woike
Sep 2022 – Aug 2023 Trieste, Italy	**The Abdus Salam International Centre for Theoretical Physics (ICTP)** Postgraduate Diploma in High Energy Physics Thesis: 2D Topological Field Theories Advisor: Prof. Pavel Putrov
Sep 2018 – Jun 2022 Giza, Egypt	**University of Science and Technology at Zewail City** B.Sc. in Physics, Highest Honors GPA: 3.87/4.0 Thesis: Spherically Symmetric Solutions in Modified Teleparallel Gravity Advisor: Prof. Alexey Golovnev

Research Experience

Oct 2025 – Present Dijon, France	**Institut de Mathématiques de Bourgogne (IMB)** Doctoral Researcher Advisor: Prof. Peter Schauenburg Topic: Orthogonality for group-theoretical fusion categories
Sep 2021 – Nov 2021 Dubna, Russia	**Bogoliubov Laboratory of Theoretical Physics (JINR)** Research Trainee (Numerical Methods) Advisor: Prof. Yakov Shnir Project: Numerical construction of topological solitons in scalar field theories
Aug 2021 – Sep 2021 Giza, Egypt	**Center of Fundamental Physics** Research Intern (General Relativity) Advisor: Prof. Shaaban Khalil Project: Symmetries in Einstein field equations for the derivation of the Schwarzschild and Kerr metrics

Talks & Seminars

Invited and Contributed Talks

12 May 2026	Modular Categories and Construction of WRT Invariants (Notes) Topology Between the Waves VI, Fréjus
7 May 2026	Positive Geometry Working Group on Geometry & Physics, IMB
10 Apr 2026	The WRT 3d-TQFT from Modular Fusion Categories Working Group on (Non-)Semisimple TQFTs, IMB
10 Mar 2026	Introduction to Quantum Groups PhD Seminar, IMB

Seminar Organization

2026 – Present	Organizer, Working Group on Stable Homotopy Theory and Higher Algebra, IMB

Selected Schools & Conferences

Aug – Sep 2026	CRC 1624 School: Higher Structures, Moduli Spaces & Integrability, Hamburg
Jun 2026	GCS '26 School: Global Categorical Symmetries, Institut Henri Poincaré, Paris
Jun 2026	Masterclass: Seiberg–Witten Theory and 4-Manifolds, University of Copenhagen
May 2026	Spring School: Khovanov and Floer Homotopy, Nantes Université, France
May 2026	Topology Between the Waves VI, Fréjus, France
Mar 2026	UNIVERSE+ School: Exploring Positive Geometry, Les Houches, France
Dec 2025	Masterclass: Shifted Symplectic Structures, Centre Henri Lebesgue, Angers
Dec 2024	Masterclass: Graph Complexes, Centre Henri Lebesgue, Angers

Teaching Experience

Oct 2023 – Jun 2024 Giza, Egypt	**Department of Applied Mathematics, Zewail City** Teaching Assistant Courses: Ordinary Differential Equations, Partial Differential Equations, Complex Analysis, Discrete Mathematics, Numerical Analysis

Awards & Honors

2024 – 2025	EIPHI Merit-Based Scholarship (M2), Université Bourgogne Europe
2024 – 2025	Tahtawi Master 2 Scholarship, French Embassy – Institut français d'Égypte
2022 – 2023	ICTP Postgraduate Diploma Scholarship (full scholarship and travel grant)
2018 – 2022	Full Undergraduate Scholarship, National Bank of Egypt (Al Ahly Bank)
2018 – 2022	Provost's Honor Roll, Zewail City (8 consecutive semesters, GPA > 3.7)
Dec 2020	Finalist, XI International Natural Sciences Tournament, Moscow, Russia
Sep 2016	National Finalist, International Physics Olympiad (IPhO), Nile University
2016	Micro Research Grant (MRG), Bibliotheca Alexandrina

Skills

Languages	Arabic (native), English (C1), French (B1)
Computational	Python, C++, Mathematica, ROOT, LaTeX